
Machine Behavior and Human Action

Speech transcribers, signature-making machines, a machine for writing personalized script, spell checkers, and bibliographic aids—these are the kinds of machines that can help humans write. They are among the behavers that humans use to assist their actions. After examining writing machines, we move on to a typology of machines in general.

Writing Machines

Speech Transcribers

For years we have been promised that the arrival of speech transcribers is imminent. These are machines into which one can speak and have one's spoken words presented in typed form on one's computer screen. For various reasons, the promise has been fulfilled only partially, and the horizon over which we expect to see a fully functional speech transcriber appear recedes as fast as we move toward it.¹ But suppose a fully functioning speech transcriber arrived on the scene tomorrow. How would we use it?

Let us suppose that the transcriber included a grammar checker and a spell checker so that it transformed our messy speech, with all its ahs and uhms, into correctly spelled, grammatically immaculate prose.² Let us compare such a speech transcriber with a skilled secretary.

1. For an explanation of this in terms of the nature of the action of speaking see, for example, Collins (1993).

2. We will return to the problems of spell checkers and grammar checkers later in this chapter.

A secretary can understand phrases such as the following: “Go back to that sentence where I mentioned the papers and ask for an additional copy of the one on solar neutrinos.” The secretary knows that this is not a piece of speech to be transcribed, but is an instruction for making a transformation of earlier transcribed text. The secretary can find the relevant passage and make the transformation even though the form of transformation is not specified in detail. The secretary will understand the meaning of “additional copy” as “an extra copy” or “a copy of an additional paper,” depending on whether the solar neutrino paper has already been mentioned. In short, the secretary is capable of acting with context-sensitive understanding of the text. A speech transcriber will have to do much more than transcribe words in order to be as useful a dictating machine as a secretary.

There are other complexities to the interaction between writing and speech. For at least some writers, the process of writing on a word processor is integral to the process of composition. If the keyboard were to be replaced by speech transcription, the speech transcription tool would, like the human secretary, have to be able to understand when it was meant to transcribe and when it was meant to understand the words as a set of instructions. In addition, it would have to understand and obey a far wider range of instructions than the secretary, and with much greater speed of reaction. The reactions would have to be fast enough to allow several versions of a sentence or paragraph to be tried and rejected at little less than thinking speed.

The speech transcribing tool would, then, have to be able to replace words, replace sections, insert sections, and so forth, by recognizing and following a set of spoken instructions that would have to be no more trouble to utter than is involved in normal backspacing and retyping—and backspacing and retyping is quite easy. It would be nice not to have to sit in one place to do the work, and not to have to maintain one’s body and arms stationary in such an awkward, upright position; it would be nice to be able to pace around the room while one’s thoughts appeared on the screen, but not at the expense of continually repeating oneself in order to explain exactly how one wanted the last passage reshaped.³

3. For bad typists, speech transcribers that do a reasonable job and then allow for easy correction of the text at a later stage might still be better than typing from scratch.

Because writing is so complicated, a careful examination of action is necessary if the relationship between humans and machines is to be understood. The polymorphic actions, which cannot be mimicked by machines without alteration of the action, and the mimeomorphic actions, which can, are deeply intermingled in writing. In the case of writing, we tend to think that mimeomorphicity goes higher up the action tree than it does.

Lettering

The ultimate transformation of fully edited words into their form as marks on paper is lower down the action tree. Let us call this “lettering.” Even here the actions are sufficiently involved with social institutions to make the substitution of human action with machine behavior much less straightforward than it first seems. Not all lettering consists of mimeomorphic actions—illuminated capitals and illuminated calligraphy rest on polymorphic actions, and we should not expect them to be straightforwardly replaceable by machine behavior.

Even the mimeomorphicity of an action is not a *sufficient* criterion for replacement by machine; it depends upon the social institutions within which the action is embedded. For example, there are sports, such as golf or high-board diving, that depend upon the execution of mimeomorphic actions, and yet replacing the humans by machines in these cases would be to miss the point of the sport. In lettering the same applies: Many cases of the production of mimeomorphic lettering are not candidates for replacement by machine. Calligraphy is sportlike in this respect. Different computer fonts on our computer are in some respects like the different hands or calligraphic scripts that can be mastered by the adult. And yet we do not admire people for their ability to use different fonts, whereas we do applaud or admire the versatile calligrapher. Of course, there is a residue of polymorphic action to be admired in the creative calligrapher, but the point we are making here is that even the mimeomorphic element of calligraphy can be admired when executed by a skillful human. As we explained in the chapter on human competence, there is no straightforward relationship between the value we put on skills and their morphicity.

The importance of the method of production of mimeomorphic script is again clear in those cases where typescript is not an acceptable substi-

tute for handwriting. To send someone a typed love letter would be odd. To send a note of condolence in something other than handwriting would be equally odd. An interesting case encountered by one of the authors concerned a note sent by a university faculty dean informing personnel of the death of a staff member; the note was laser-printed with a font that mimicked human handwriting. One can see what the dean was trying to do, but the fact that the note was not handwritten was taken by many staff to indicate a failure on the part of the dean to understand what we might call “the aesthetics of emotion.” Strangely, a handwritten note that had been obviously photocopied would have been more acceptable.

There are opposite cases where writers are expected to avail themselves of machines. Because of its relative illegibility, to distribute a scientific paper to one’s peers in handwritten form would nowadays be judged impolite, and to send such a thing to a journal editor would be inappropriate.

The Signature

Another case where machines can replace the pen in only special circumstances is the signature. Making a signature, a quintessentially (self-copying) mimeomorphic act, is a mark of identification; just as bullets show the marks of the gun that fired them, and typewriters have their specific quirks that enable the police to work out which machine was used to type a threatening note, the written signature is, to use the word metaphorically, the “signature” of a particular biological machine—you. Those who forge signatures disguise their biological machinery as that of another. In the normal way, then, it would be inappropriate to use a machine to make your signature, because the test of identity works only if it is the biological you who utters the signature.

One exception to this rule is the signature machines used by banks and other large financial institutions; another is the signature on a banknote; another is the printed facsimiles of signatures to be found on advertising junk mail. In these cases, however, we are not really looking at signatures but at signature tokens (again, “token” is not used here in the sense of “action token”). In the case of the banknote, the exact form of the signature is of no interest to anyone—it must be one of the easiest elements of the banknote to forge. In the case of signature machines, we

are substituting the “signature” of a metal mechanism for the signature of the biological machine; it would not matter if the signature produced by the machine exactly matched the signature of the corresponding human, for the signature indicates only that the check was written within the institution that housed the signature machine. The signature machine, then, fulfills the same role as an official seal, rather than that of a human being. In the case of junk mail, it is rather hard to see what the printed signature is for; perhaps some take it as a symbol of personal attention, but surely only the most gullible. In the case of a typed letter, the sender is expected to sign by hand if they mean to show that it has been composed—or at least read—by the sender.

On the other hand, one of the authors once worked in the office of an American senator who owned a different type of signature machine. This was a record-and-playback device into which an ordinary pen could be fixed. If a letter was properly positioned, the metal arm of the device would move the pen in such a way as to reproduce the senator’s preprogrammed signature. Here the intention was to deceive the receiver of the letter into thinking that the senator had given more personal attention to the response than was in fact the case. This machine was kept well hidden from visitors to the office, reinforcing the point about the norms associated with signing letters. We see here a machine used to effect a deceit—a case of an inauthentically executed mimeomorphic action. It is interesting that even mimeomorphic actions can be inauthentic. Thus we see that the distinction between authenticity and inauthenticity does not map onto the distinction between polymorphic and mimeomorphic actions.

Is There a Market for a New Kind of Writing Machine?

The senator’s signature machine, along with our analysis of lettering, invites the invention of a new kind of word-processor program. Such a machine would learn the character of a writer’s hand, through the input of various samples of writing, with the aim of being able to develop a font that reproduces the hand (with various haphazard variations), in response to keyboard input. Allied with a suitable color printer (or even a pen manipulator), this device would produce text that could be hard to distinguish from the handwriting of the typist. A machine such as this

would be useful to someone who finds typing much easier than handwriting, or who finds the keyboard and screen an indispensable element in composition, but who still wants to write personal notes and letters. Admittedly, sending this kind of letter to strangers would involve an element of deceit, while sending it to friends might suggest, like the note sent by the dean, a lack of aesthetic judgment. It is possible, however, that such machines would change the way we think about handwriting. The reception of such a program would make a nice case study in the meaning of “personal” within our society.

Spell Checkers and Bibliography Checkers

We now travel back up the writing tree a little way to discover another feature of the relationship between polymorphic and mimeomorphic in the action of writing. Consider spell checkers. Even spelling can be a polymorphic action. Occasionally we want to be adventurous in our spelling, as in our love letter examples or in poetry or some kinds of novel. More problematically, consider that a spell checker will correct the misspelling in *antidisestablishmentarianism* (a mistake that many humans would miss), and in *weerd* processor, but not in *world* processor (which nearly every human would spot). This is because *world* is correctly spelled though *weerd* is not. More complicated still, *antidisestablishmentarianism*, *weerd* processor, and *world* processor are all spelled just as we want them in this passage of text as it appears in this book, so any corrections the spell checker made would be mistakes. The spell checker would have to understand the whole sense of this passage if it were to spell as well as the typical editor. Thus we value a degree of polymorphic in our spelling, and because of this it is currently impossible to design a spell checker that checks in the same way a human checks.

Compare spelling with entries in the bibliography of an academic book or paper. When it comes to bibliographies, we seem to be interested in standardization and uniformity and not at all concerned with creativity, freedom of expression, or context dependency (except of a very narrow, disjunctive-mimeomorphic sort—i.e., which style for which journal). It would be very unusual for someone to claim that their work is especially valuable because of its sensitive bibliographic response to social circum-

stance. Thus in the case of bibliographies we do not seem to value polymorphicity at all; it is all mimeomorphicity.

Here is a typical bibliographic warrant offered by a sensitive writer, Steven Shapin (1994, p. xxiii):

This book uses an economical footnoting and referencing convention identical to that employed by Shapin and Schaffer's *Leviathan and the Air-Pump* and similar to that of Elizabeth Eisenstein's *The Printing Press as an Agent of Change*.

As can be seen, the stress is on convention and similarity to previous usage by self and others, whereas such an author is normally concerned to emphasize creative features of his or her texts. An author who wrote that his or her referencing system was idiosyncratic and should not be adopted by others would not have understood the point of referencing as it is valued in our society. For this reason we would predict that referencing conventions will converge and that eventually referencing that is not aided by machine will become rare. We predict that references will come to be stored in standard databases, that there will be programs that format them for different house styles, that the number of house styles will rapidly diminish, and that all these changes will be welcomed by almost everyone, and will *not* be seen as an example of the growing tyranny of machines, even by "neo-Luddites." This, of course, is not a particularly adventurous prediction since the trend is well under way.

What Machines Can Do

Some of these complexities can be understood within a more general framework. What kinds of machines are there in the world? There are at least two ways of typologizing machines, the first is according to what they do, and the second is according to how they work. We deal first with what machines do.

Machines are of three kinds:

1. *Tools*: Machines can amplify our ability to do what we can already do—let us call such machines "tools." A hammer is a tool; a mechanical digger is a tool; a word processor is a tool; a medical diagnostic help system is a tool.
2. *Proxies*: Machines can replace us by doing what we already do (perhaps better than we can do it)—let us call such machines "proxies."

A rope tied round a bollard to hold a boat still is a proxy; a thermostat is a proxy; the control belonging to a radar-controlled gun is a proxy; a computer program that integrates mathematical functions is a proxy; an expert system for medical diagnosis is a proxy.

3. *Novelties*: Machines can do types of things that we could never do without them—let us call such machines “novelties.” A high bridge is a novelty; an explosive device is a novelty; a fridge-freezer is a novelty; a space rocket is a novelty; a laser is a novelty; a virtual-reality headset is a novelty.

The boundaries between the three categories are, as we have come to expect of these things, less than watertight. The differences are partly a matter of the level of analysis. Consider first the difference between tools and proxies. Between the moment that one presses forward the little lever on a tool such as a mechanical digger, to instruct it to drag the bucket, teeth first, through the earth toward you, and the moment the bucket comes to the end of its travel, the device is acting not as a tool but as a proxy—it is scooping up earth “by itself.” When action is broken down into small enough elements, tools become proxies. To take an example from our analysis of writing machines, if one is word processing a document that one intends to present as typescript, from the moment one completes the “print” instruction to the moment the document emerges in the paper tray, the word processor/printer is a proxy; it is typing for one.

On the other hand, we can always turn a proxy into a tool by considering what it does from a vantage point higher up in the action tree—thus, a rope is a tool for our ship-controlling actions, a radar-controlled gun is a tool we use in our war-waging actions, and a word processor is a tool we use in our writing actions.

The general rule is that we can always turn a tool into a proxy by taking a lower vantage point, and we can always turn a proxy into a tool by taking a higher vantage point.

Novelties are only a little more robust. A space rocket can be seen as a tool by standing at a very high level of the tree (it amplifies our ability to travel), and even into a proxy with sufficient determination (it travels for us?). A fridge-freezer, which seems at first sight to be able to do something that no human can do (freeze water), can be said to amplify our ability to preserve food or even, attributing agency, to preserve food

for us. Likewise, tools can be seen as novelties (the digger bucket lifts a ton at a time), and so can proxies (a rope can hold a huge ship for ages and ages). Our word processors are innovators, to the extent that they enable us to integrate creative composition and the production of script to an unprecedented extent.

What we count as a tool, a proxy, or a novelty depends then upon our standpoint in the action tree. But this “relativity” rests on something that is not so relative: if correct description of a machine depends on standpoint, there are also incorrect descriptions of machines where the relationship between standpoint and attribution of degree of independence is incorrect. The burden of much of the argument of this book and of previous works is that what have been taken to be proxies should really be thought of as tools. It is easy to mistake tools for proxies because of the ease with which we “repair” the deficiencies of machines and then “attribute” agency to them.⁴ This is what we will call “Repair, Attribution, and all That” or “RAT.”⁵

RAT: Hiding It, Balancing It, Killing It

Pocket calculators do not “do arithmetic”; they do only a tiny part of arithmetic. We think a pocket calculator does arithmetic after the fashion of a proxy, or even an agent, because we continually repair its errors and attribute high-level arithmetical agency to it. If you pick up a cheap pocket calculator and do the sum “ $7/11 \times 11$ ” you get the answer 6.9999996, but you do not immediately write a letter to the manufacturers telling them that they have sold you a dud calculator. You look at the result and you “see” 7. Seeing 7 where there is only 6.9999996 is part of the work you do in making the interaction between you and the

4. We say nothing here of the morality of the attribution of these terms to machines. Ask the question of the senator’s signature machine. Was it a proxy for the senator? One might say “yes” in that it reproduced the senator’s signature for him, or one might say “no” because a machine cannot replace a senator in his role as one who distributes his limited personal attention according to the deserts of the cause. The signature machine appears to increase the total of the senator’s personal attention; on the other hand, the value of personal attention would seem to be commensurate with its scarcity.

5. See Collins (1996).

calculator look like ordinary arithmetic. If one analyses the process more carefully, one finds that a lot of work is done by the user, both at the input stage and at the output stage.⁶

This work is of the same sort as is needed when one hears the strange accents and intonations of a voice-generating chip uttering a telephone number, or insisting that “a door is a jar” (a phrase frequently evinced by some cars built in the 1980s).⁷ Users can turn mangled vocalizations into English words because they can provide enough context to make sense of what such a device is “trying to say.” On the input side, voice-operated computers require users to speak in a strange stilted way if they are to work. This is because they *cannot* do the same sort of interpretative, contextualizing work on humans’ varying vocalizations that we can do on theirs. The special way of speaking that users have to adopt is equivalent to their being ready to insert an arithmetical sum into the calculator through a keyboard that offers a limited choice of options in the way the sum is expressed.

What one is doing when one sees 6.9999996 as 7 and hears the voice chip as saying ordinary words is known as “repair.” The user is repairing the deficiencies of the computer. Insofar as we do such repair work without noticing it, we can easily think of the calculator (voice chip) as doing arithmetic (speaking). It is rather like the process of anthropomorphization. People might attribute human characteristics to their dog, to their car, and to their calculator. This is what we are doing when we say that our calculator is “better at arithmetic than we are.” We are doing repair and attribution and all that.⁸

“Hiding the RAT” is the most general principle of computer interface design. The interface is designed so that the user is unaware of how much RAT he or she is putting into the interaction. The way to do this is to have the computer need only the kind of RAT that we are accustomed to using in our everyday lives. In other words, make the interface require

6. See Collins (1990), chap. 5; and Collins (1997).

7. A very well known English riddle goes “When is a door not a door?—When it’s ajar” [a jar].

8. The source of the idea of “repair” is the ethnomethodological literature. See, for example, Garfinkel (1967).

only ubiquitous skills. This is called making interfaces “user friendly.” It is less a matter of making the interface do skillful things and more a matter of encouraging users to take advantage of the skills they already have.

The examples given in the previous section are like this. Every pre-calculator era arithmetician was already accustomed to approximating, because the logarithm tables and slide rules that they used required even more approximating skills than do calculators. That is why the calculator’s mistakes are so easy to repair, and why it is so easy to attribute arithmetician’s skills to them. Ubiquitous skills are transparent; one does not notice that one has them. The repair of broken speech patterns is another ubiquitous skill. One of the things that designers are doing, as programs become more user friendly, is to make their programs touch our culture in places where the skills of repair are already in place (Collins 1990, chap. 7).

One design mistake is to think that a computer that thoroughly hides the RAT does not need RAT at all. This kind of mistake led, for example, to the overblown claims for the power of expert systems such as automated medical diagnosticians—for example, MYCIN. Such mistakes happen when computer enthusiasts fail to understand how much work and skill humans put into their day-to-day interactions and into their interactions with machines. If one does not notice how skill-laden is ordinary social life, one may think that a system with which people can interact easily is a social creature. But it is not a social creature unless the pattern of skill use is roughly symmetrical between all parties. Once again, the asymmetry is easily seen in systems that involve speech, but careful analysis will show its presence in nearly all computer-human interactions.⁹ Only in special cases is symmetry achievable with current or immediately foreseeable machines.

To move the correct attribution of agency up the tree, the RAT has to be “balanced.” At the top of the tree, human interaction, and human-machine interaction, involve a lot of interpretation and repair. “Balancing the RAT” means building devices that are as good at RAT as are humans,

9. For a hilarious account of “unbalanced RAT” (our term) in automated photocopiers, see Suchman (1987).

so that the burden of repair is distributed evenly between both parties in an interaction, even when one is a machine. In current computer-human interactions nearly all the RAT is done by the humans.¹⁰ Only if the RAT is balanced could the machine be said to be taking the place of a socialized human being working at the top end of the action tree. Only a conversationalist that could balance the RAT could pass a Turing test in which the interrogator asked questions (in misspelled English full of puns and so forth) that required for their rectification the sort of understanding gained through socialization. No ordinary balanced-RAT interchange can be managed unless, one way or another, we put social knowledge into computers.

There is another way of balancing the RAT; this is to eliminate the need for it—what one might call “killing the RAT.” Machines can take the role of ordinary, socialized, human beings without being members of society under some circumstances. The circumstances are that ordinary humans beings want to do things that could be done without the skills, ways of conceptualizing the world, and ways of acting and speaking that are normally mastered with socialization. This way of doing things is, of course, mimeomorphic action.

The Proper Attribution of Agency

To paraphrase the earlier argument about calculators in terms of the ideas set out above, we can say that agency is correctly attributed to calculators only when the vantage point is low in the action tree. Careless optimists, however, continue to talk and think of calculators as proxies, or even agents, even when they are seeing them from much higher in the tree—the attribution of agency “slips up the tree” as it were.¹¹ Careless optimists think they are seeing high-level agents when they are seeing only low-level proxies, and they maintain the illusion by repair and attribution.

10. This is not to say that computers and other machines cannot do various bits of repair such as elementary spell checking, but as we have explained above, they fall well short of human competence because of the polymorphic element of spelling.

11. It could be said that this is what Drew McDermott (1981) was getting at in his article, “Artificial Intelligence and Natural Stupidity.”

Other supposed proxies simply do not work because their users are unable to supply the necessary amount of RAT. Thus there simply are no satisfactory medical diagnostic expert systems, only diagnostic help systems.¹² Help systems are meant to aid skilled users, whereas expert systems are meant to replace expert users. Installing an expert system within a culture that repairs its errors defeats the object—it becomes a help system and is better designed with that goal in mind. Furthermore, in the circumstances of most hospitals, the mistakes of diagnostic expert systems are readily detected precisely because the systems are monitored by skillful doctors. To find circumstances in which a diagnostic system will work as intended, one must find a location where it is the only resource. Hartland (1993) found that a heart-disease system was never used in city hospitals but was used in a remote rural hospital to refer patients to specialists, because at the rural hospital a high level of “false positives” (with the associated worry and waste of resources) was more acceptable than a small number of false negatives (with fatal consequences), and there was no skilled doctor on hand able to do better.

Machines as Proxies

In this book we are interested in machines as proxies. We have almost nothing to say about novelties, and little to say about tools except that they are often mistaken for proxies. We are interested, in short, in where current and foreseeable machines can replace us—where they can do what we are already doing. And our argument is that machines can only be proxies where whatever we want to bring about can be brought about through mimeomorphic actions.

We can now see one way in which this makes sense. Machines are turned from tools to proxies by going down the tree; but, of course, as we go down the tree we specify actions ever more narrowly, such that they tend toward mimeomorphicity. There is less “relativity” than we first thought! More narrowly specified elements of the tree tend to be mimeomorphic and therefore tend to be more available to replacement by mechanical proxies.

12. See Lipscombe (1989). Or systems that work because the organizations adjust to them in various ways (Berg 1997).

The section on writing machines shows that even this relationship is not straightforward. We discovered that speech transcription, as carried out by a human secretary, contained polymorphic actions. We found that typing, when it was an element within creative composition, was intimately tied up with polymorphic activities. We found that spelling could contain a polymorphic element. We found that a person's signature, quintessentially mimeomorphic though it is, could not be reproduced by a machine without turning into something like a "seal of office," or becoming a kind of forgery (as where the senator used his signing machine to give the impression that his own biological mechanism had produced the mark). These, however, are caveats and qualifications rather than councils of despair. A speech transcriber could be a reasonable proxy for a copy-typist, and in spite of the problem with spell checkers, automated bibliography editors can be expected to do their job increasingly well as proxies for *part* of the job of a human editor. We can also see that spell checkers and speech transcribers would make good tools so long as their limitations are understood—they can amplify and improve our abilities to do things.

There is, of course, another way to make mimeomorphicity acceptable higher up the tree. This is to change our forms of life. If, say, we begin to speak after the fashion of George Orwell's *1984*, we could have mechanical speaking proxies just as we now have mechanical telephone number-uttering machines. Speech transcribers and spell checkers would then do a much better job as proxies rather than tools.

Our story is about the ultimate limits of proper attribution of agency to machines. How far up the tree can we go? Our argument is simply that we cannot see how to properly extend agency to machines beyond the point in the tree where polymorphic actions begin.

How Machines Work

Doubtless, as machines develop, their potential for use as proxies creeps up the tree a bit, even if everything else we do remains the same. How much difference does the improvement in mechanism make? Previous analyses of types of machine generally talk of devices "acting." Action, as we have stressed, however, is for us a term that connotes intention, and machines do not have intentions. In our terminology, machines "behave"; machines, then, are "behavers."

Bright (1958) divided machines up into seventeen types, but for our purposes we need four:

Behavers: Behavers just do the same thing over and over again. An automatic drilling machine is an example of a behavior.

Disjunctive behavers: Disjunctive behavers do one of a set choice of things according to a predetermined list of choices. A chair-spraying robot, programmed to recognize which of a set of types of chair is currently in front of it, and to operate the corresponding program, is a disjunctive behavior.

Feedback behavers: Feedback behavers respond not to a fixed set of possibilities but to anything within a preconceived *range* of sensed stimuli. A car wash is a feedback behavior—it senses the shape of the car and adjusts the position of the rollers accordingly for any shape of car so long as it is within a preset range (and doesn't have a radio aerial or too many other loose bits). Note that the behavior of a feedback behavior is not predictable in advance; whenever a manufacturer decides to produce a new model of car, a car wash encountering the new model will follow a sequence of movement that was impossible for the car-wash designers to predict.

Learning behavers: Learning behaviors take in information from their environment and their users, or both, and incorporate this in their design. Examples include plaster of Paris casts such as are used for holding broken bones in place; the cast takes in information about the shape of the leg, then sets in a shape that models the limb. A pair of good heavy leather boots is similar—they “learn” the shape of the foot—though with boots the learning process is never quite complete. In the same class are rule-inducing expert systems, which establish the “shape” of their program through comparing examples and using feedback from their “trainers,” and, of course, neural nets, which carry the process further. Most rule-inducing expert systems and neural nets are much more like plaster casts than heavy boots, because they are “set” after the learning period is complete. Some expert systems (such as the crystal-growing system described in Collins [1990]), can continue to learn from their users, however, and this, no doubt, is the aim of neural net designers.

Interim Summary

The argument so far can be summed up diagrammatically (see figure 7.1). Here tools, proxies, and novelties are represented by triangles, while human action is represented by a rectangle of polymorphic actions

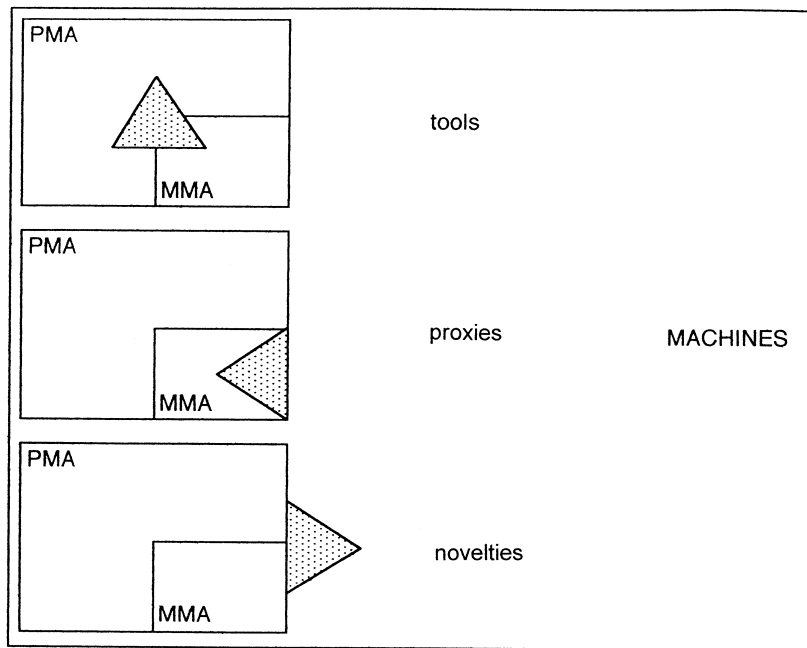


Figure 7.1
The territories of machines and actions.

enclosing a rectangle of mimeomorphic action. Tools meld into the world of polymorphic actions (though close examination from a vantage point low in the tree would reveal them to be proxies, managing the mimeomorphic elements of polymorphic actions). Novelties extend our abilities, and, thought of as tools from a high level of the tree, they amplify our abilities. (It is easy to see that a hammer can be thought of as a tool or as a novelty.) Thus in the diagram, novelties are shown extending out to the right beyond the domain of human action and into the unknown. Proxies are invading the inner rectangle and taking it over (usually to our great relief and pleasure). Proxies do better and better as they gain abilities such as learning from feedback; they push into the domain of mimeomorphic action.¹³ On this diagram, the only difference between a

13. In Bright's (1958) account, people working in factories needed less and less skill as proxies invaded the whole of our territory; on our account, and the account of many of those who have carried out "studies of work," factory

canoeing machine and a bicycle-balancing machine is that the first has not yet been developed, and the second has.¹⁴

It may be that humans too are engaged in their own invasion. Here and there perhaps, humans are invading the territory of novelties. Through studies of diet and training regimes, athletes are getting stronger and faster, learning how to do things that once only machines could do. Perhaps, with careful use of biofeedback, abuse of drugs, and interference with our genes, we humans will one day teach ourselves how to do what only a fridge-freezer can do today; we may one day be able to freeze ice with our bare hands! But if we do, we will probably be able to do it in two ways, involving polymorphic actions, or mimeomorphic actions. Machines—at least, the ones we understand today—will still only be able to mimic mimeomorphic water freezing.

The Phylogenetic Scale of Machines

It is tempting to think that as machines begin to learn from feedback, they are approaching us in ability. Where should we put machines on the phylogenetic scale of living organisms? Around the level of apes, perhaps? Our diagram is meant to suggest that the invasion of proxies will not cross the boundary of the polymorphic actions rectangle. In part, our choice of examples was also made to prevent too optimistic a judgment. While a neural net can be thought of as “almost human” because of the way it “learns” from feedback, “works out” its own rules, and “remembers” them, it is less tempting to impute intelligence to a car wash or a pair of old boots.¹⁵ Why is this? The answer seems to have something to do with what we count as one of the high-water marks of human ability.

workers, even those who attend highly automated machines, still need all the skills associated with the contents of the polymorphic actions rectangle. Bright (1958), is quoted in Braverman (1974, p. 216). This point will be discussed at greater length in the chapter on bureaucracies and mechanistic systems. We are grateful to Jamie Fleck for drawing this material to our attention.

14. And this, to reiterate a point, is why the term “situated action,” which covers both canoeing and socially situated actions, does not fit neatly into our scheme.

15. Though Herbert Simon (1969) seems ready to impute intelligence to a thermostat.

Once upon a time, the ability to calculate was taken as the true stamp of cleverness. The invention of computers has done something to lower this ability in our estimation. Nevertheless, while calculation is no longer a high point, it does seem to be an essential ability for those entities to which we attribute intelligence; an old boot does not calculate.

On the other hand, a simple program such as ELIZA does its work through calculation and it must be this, in part, that makes it such a tempting candidate for charitable assessment.¹⁶ So long as acting competently *in the flux of social life* is not taken as the quintessential mark of human ability, we will think of ourselves as complicated ELIZAs, and ELIZA will continue to look good. Thus the typology of the ways in which machines work, set out above, if it is a phylogenetic scale, has a much narrower range than first appears. If we can establish the crucial importance of social abilities, we will have shown that even the most advanced neural nets are at the level of the mollusk. We will return to this theme in the conclusion.

Indifference Matching: The Sociology of the Same

The typology of the ways in which machines work matches our typology of human mimeomorphic action. Ordinary mimeomorphic actions can be replaced by behaviors, and the various kinds of disjunctive mimeomorphic actions can be replaced by disjunctive, feedback, and learning behaviors. These kinds of machines have the potential to be our proxies under the appropriate circumstances. But what are these circumstances? It is a matter of “the sociology of the same.”

The whole pattern of human life could be said to be a matter of what we see as the same and the way boundaries between things are shifted about. Artists, scientists, politicians, writers, and all the others who help to change our culture are all the time trying to establish new relations of similarity and difference. To return to the more mundane example discussed in an earlier chapter, what makes golfers golfers is that two 300-yard drives are seen as the same if they follow a similar flight path and land within a few feet of each other, whereas what makes dog owners

16. See Collins (1990).

dog owners is that they see those two drives as different if one hits a dog and the other does not; what makes engineers engineers is the tendency to see the two golf swings as different in terms of the scale on which engineers usually measure things. The golfer, as golfer, is indifferent to the differences seen by the engineer and the dog owner. We can happily use machines as our proxies when the envelope of our indifference to the way we execute our actions is no narrower than the envelope of variation in the behavior of the machine. It happens, then, that we can build machines that can drive golf balls with less variation than that of the best golfer—one such is known as “Iron Byron.” It is a fine golf ball-driving machine. Indeed, since even the best golfers would prefer to drive with less variation than they can actually achieve, they count Iron Byron as a better driver than themselves.

There are, on the other hand, no machines that could replace golfers in the whole of the game.¹⁷ That is because our envelopes of indifference for how the whole game is played are relatively narrow. The action “playing a game of golf” is full of structure that we care about deeply. To get to areas of indifference within which a machine or machines could fit, we have to go well down the tree to aspects of the game such as driving.

Of course, the structure within a triangle of action can be matched by a machine so long as it is a disjunctive structure—a matter of specifiable relations and functions—but the whole game of golf is not like this.

Let us explore the matter in the context of another of our standard examples. Consider the way a pocket calculator is made to fit into our form of life. Consider especially the example of $7/11 \times 11$ when the calculator gives 6.99996 as the answer. What we do when that happens is to repair the calculator’s problem—we say “it really means seven,” so it is as though the calculator has never done anything wrong at all. Now consider the sum “ $7/2 \times 2$.” Most calculators will give the answer “7” to this sum, which we count as the correct answer. But suppose I happen to have a calculator that gives the answer “7.” Do I “repair” the

17. We ignore the whimsy inherent in trying to replace a human golfer with a machine—that of course would destroy the point of the game as a test of human skills. We are talking here of in-principle, engineering feasibility.

calculator's answer? Does the question make sense? At this point the reader may be thinking that the copy editor has made a mistake. The reason the question does not seem to make sense is that there is nothing to repair. The answer "7" and "7" are different but not in a way that is in need of repair as far as arithmetic is concerned. We are indifferent to the difference between 7 and 7 in a way that we were not indifferent to the difference between 7 and 6.99996. Or to put it another way, our indifference to the difference between 7 and 6.99996 is an active indifference—it requires us to do something positive in the way of repairing so that the difference does not count anymore, whereas our indifference to 7 and 7 is passive.

That passivity does not have to do with the magnitude of the difference between, on the one hand, 7 and 7 and, on the other hand, 7 and 6.99996. There is no absolute magnitude to these things. It has to do with the relationship of the numbers to the form of life of arithmetic. The way we cut things up in arithmetic makes 7 and 7 "the same" while 7 and 6.99996 are different unless we do something about it. But we could imagine it otherwise. We could imagine that the difference between ordinary and bold print comes to have an arithmetical significance. In that case, the bold 7, if it was the wrong answer, would need repair in the same way as 6.99996 does now.

The importance of these examples is to show that while the theory of the shape of actions (the theory of action morphicity) is a human-centered theory, as are theories of repair or attribution, the center is in a different place. Repair or attribution theories account for every feature of a mechanism, such as a calculator, by reference to users' interactions with it. The theory of action morphicity accounts for some features of such a mechanism by reference to the interactions of users among themselves, at a point logically prior to their encounter with the mechanism. The first type of theory says that whatever the mechanism is seen to be is a matter of the way the users see it; the theory of action morphicity says that whether a mechanism will fit into a form of life without repair depends on the form of life. A mechanism may be made to fit better or worse by changing the form of life. How much repair work needs to be done is a matter of the extent to which form of life and mechanism match. They may come to match more closely if the form of life comes to contain

more mimeomorphic actions. They may be either *casual mimeomorphic actions*, so that we are indifferent to more things, or *special mimeomorphic actions*, so that where machines produce what we count as constancy of outcome we are especially pleased. When machines act as proxies in replacing casual mimeomorphic actions, we like it because it saves us time and energy; when machines replace special mimeomorphic actions we like it even more because, usually, we consider that the machines do a better and more accurate job than ourselves. Machines are usually better than us at generating behaviors equivalent to special mimeomorphic actions—good machines are made that way.

There are several *sociologies of the same*. There is the sociology of how new relationships of sameness are established—this is what the sociology of scientific knowledge is about. Studies of “scientific discovery” show us developing new ranges of indifference to variation; a laser that works is “the same as” another laser that works irrespective of the multitudinous small differences of construction; a laser that does not work is “different” to one that does work even if there are no differences apparent to the eye.

Another sociology has to do with the ordinary application of the word “same” to broad classes of entities; what sort of thing do we associate with constancy. For modern Westerners, the idea that we can never step into the same river twice comes as a bit of a shock, because the scientific viewpoint, as opposed to, say, the magical, makes nature a hallmark of constancy and predictability. Machines, too, are thought of as reproducers of constancy and predictability. Detailed studies of machines, such as Kusterer (1978), which show them to vary markedly from individual model to individual model, again disturb our preconceptions. That which is human and human-sized seems to us to be variable and “rough,” while that which is mechanical and small seems to us to be constant and “smooth.” To us, a paradigm of smoothness is a polished metal surface—say the inside of the cylinder belonging to the engine of a modern car—but examined under the microscope it is a mountain range of jagged hills and valleys. All smoothness, all constancy, is relative in this way; it depends on perspective, and therefore all judgments of smoothness and constancy depend on the choice of scale. The sociology of the same is about the differing application of the terms constancy, smoothness, and

so forth, in different forms of life. What we have tried to show is that in spite of the long-term relativity of the application of such terms, it is still possible to make mistakes in applying them. If we do not, as a society, feel indifferent to certain differences, then a machine that is indifferent to those differences will not reproduce our actions satisfactorily. To call an action mimeomorphic is to describe the area within which we feel indifferent to the behavioral instantiation of the action. Indifference and similarity are the same idea; it is indifference that allows us to induce.¹⁸

Summary

The analysis of machines that might be used to help us write showed how intimate and complex is the relationship between ourselves and machines, and yet how the idea that only mimeomorphic actions can be replaced by machines robustly survives examination.

We went on to typologize the role of machines into three categories—tools, proxies, and novelties. We showed how taking different perspectives alters our concept of which of these things a machine is. We showed, nevertheless, that it is still possible to make a mistake by choosing the wrong idea about a machine from a particular perspective. The idea of RAT was used to show why the nature of machines' contribution is so easily overlooked. Repair, Attribution, and all That is such a widespread facility that we use it all the time without noticing, thus causing us to anthropomorphize mechanisms. The only kind of mechanism that should be a candidate for the label "humanlike" is one that has as much facility at RAT as ourselves. We have no such devices except where mimeomorphic actions are concerned; in this respect, Iron Byron is more humanlike than ELIZA.

We then typologized machines into different types of "behavers" and discussed the phylogenetic scale of devices. This way of looking at things showed the relationship between tough boots, plaster casts, and neural

18. Rule-inducing expert systems put the cart before the horse. Enthusiasts for such expert systems believe the systems will discover the regularities in the world, whereas it is we who decide on the regularities by deciding on the things we are indifferent to. This abstract idea has quite concrete manifestations in the failure of so-called discovery programs (see, for example, Collins 1989).

nets. We used a diagram to look at the intermingling of machines and humans in another way. Finally, we showed how the idea of indifference and the idea of mimeomorphicity were related. Thus we see what humans and machines can do.

Appendix: A Robot Slave

Suppose a robot followed its owner about, mimicking everything that he or she did? If the robot was successful, showing that every action that the person did could be mimicked by a machine, would this show that every action the person did was mimeomorphic?¹⁹

We describe machines as “mimicking” actions rather than acting, so as to make clear that entities that do not have intentions do not act. When we say that a machine mimics an action we mean that the *consequence* of the mimicry is the same as the *result* of the action. There is, however, another use of the term “mimic.” The type of robot mentioned in the query mimics the behavior, but this does not necessarily have consequences that are the same as the result of the action. This is not surprising in the case of polymorphic actions. For example, if the robot’s owner greets a secretary with a cheery “hello,” and the robot immediately repeats the greeting in the same tone of voice with the same gestures, this is likely to be seen as “mockery” rather than greeting. To copy the behavior associated with the action of greeting—which means having it carried out twice, not once—is not to mimic the action.²⁰ The same goes for *mimeomorphic* actions where a double execution does not have the same value as a single execution. Mimicry of behavior is often used as mockery by humans as well as by machines, and it matters not whether the action corresponding to the behavior is polymorphic or mimeomorphic. What we conclude is that not all copying of behavior, even in the case of mimeomorphic actions, is mimicry of the action. Nevertheless, no action can be mimicked by copying the action unless it is a mimeomorphic action.

19. This question was put to us by Olga Amsterdamska.

20. There are a number of alternative analyses possible. Perhaps a low-level action is being mimicked—e.g., “saying hello”—whereas the higher-level action—“greeting”—was not.

Now let us make the case more difficult. Going back to the polymorphic action of greeting, suppose the owner's robot, instead of following the owner around, was controlled remotely by the owner's actions. In a scenario beloved of philosophers, the robot's owner sits at home while the robot-facsimile wanders around in the world with its senses fully and instantly linked to the owner's senses via telemetry—the robot is controlled by the owner's movements—everything the owner does and says, it does and says.

Through the eyes of the robot the owner sees the secretary and cries "hello." The machine mimics the owner's behavior, but on this occasion it also has the effect of executing a greeting. On the face of it, the robot has mimicked a polymorphic action without having to be socialized first. Does the theory fall?

It does not fall, because the robot is a tool rather than a proxy. Or, to be more exact, it is a proxy only when the elements of the action are broken down into very small bits indeed. As far as the action of greeting is concerned, the robot is not doing the owner's action for the owner, merely enabling the owner to do the action better—that is, from a remote site. To think of the robot as mimicking the action of greeting is a bit like thinking of the owner's lips and mouth as mimicking the action of greeting when the owner says "hello." The owner's lips and mouth are not mimicking the action, they are helping the owner execute it. The robot is the same.